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5305

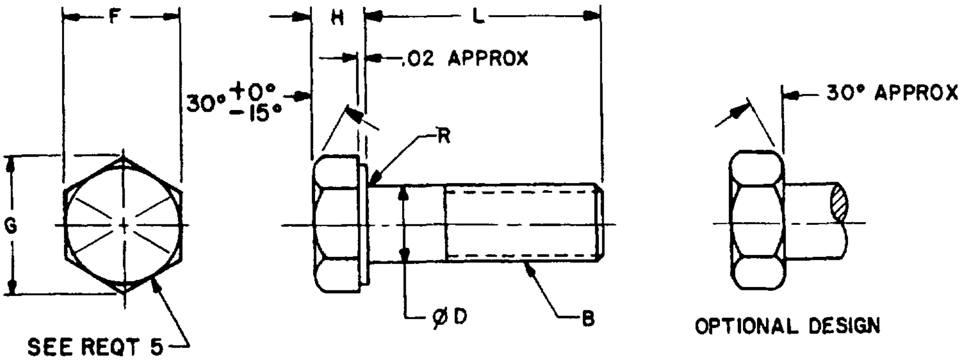

TABLE I. DIMENSIONS AND PART NUMBERS.

NOMINAL SIZE			.250	.312	.375	.438	.500	.562	.625	.750	.875	1 000	1.125	1.250
B - THREADS PER INCH UNC-2A - - - - - - - - -			20	18	16	14	13	12	11	10	9	8	7	7
Ø D - BODY DIAMETER - - - -		MAX	.2500	.3125	.3750	.4375	.5000	.5625	.6250	.7500	.8750	1.0000	1.1250	1.2500
		MIN	.2450	.3065	.3690	.4305	.4930	.5545	.6170	.7410	.8660	.9900	1.1140	1.2390
F - WIDTH ACROSS FLATS - - -		MAX	.4375	.5000	.5625	.6250	.7500	.8125	9375	1.1250	1.3125	1.5000	1.6875	1.8750
		MIN	.4280	.4890	.5510	.6120	.7360	.7980	.9220	1.1000	1.2850	1.4690	1.6310	1.8120
G - WIDTH ACROSS CORNERS - -		MAX	.5050	.5770	6500	.7220	.8660	.9380	1.0830	1.2990	1.5160	1 7320	1.9490	2.1650
		MIN	.4880	.5570	.6280	.6980	.8400	.9100	1.0510	1.2540	1.4650	1.6750	1.8590	2.0660
H - HEAD HEIGHT - - - - -		MAX	.1630	.2110	.2430	.2910	.3230	.3710	.4030	.4830	.5630	.6270	.7180	.8130
		MIN	.1500	.1950	.2260	.2720	.3020	.3480	.3780	.4550	.5310	5910	.6580	.7490
R - RADIUS		MAX	.0250	.0250	.0250	.0250	.0250	.0450	0450	.0450	0650	0950	.0950	.0950
- - - - -		MIN	.0150	.0150	.0150	.0150	.0150	.0200	.0200	.0200	.0400	.0600	.0600	.0600
1/ MIN TENSILE LOAD - LBS - - - -			3,800	6,300	9,300	12,800	17,000	21,800	27,100	40,100	55,400	72,700	80,100	101,700
L LENGTH	TOLERANCE		DASH NO	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO
	SIZE .750 AND UNDER	SIZE .875												
.500	+.000 -.031	+.000 -.062			018									
.562							039							
625					019		040							
.750				011	020									
.875			001		021		041							
1.000			002	012			042		058	072				
1.125	+.000 -.062	+.000 -.125					043							
1.250			003	013	022	035				073				
1.375							044							
1.500			004	014	023	036		057	059	074	089	101		
1.750			005		024		045		061	075				
2.000			006		025		046			076	090	102		
2.250	+.000 -.094	+.000 -.188			026		047		062	077	091			
2.500			007	015	027	037	048		063	078	092	103		
2.750					028		049		064	079				
3.000			008	016	029	038	050		065	080	093	104		
3.250					030		051			081	094			
3 500			009		031				066	082	095		108	
3.750										083				
4.000			010		032		052		067	084	096	105	109	
4.500					033		053		068	085	097	106	110	113
5.000				017	034		054		069	086	098			
5.500							055		070	087	099		111	
6.000							056		071	088	100	107	112	114

1/ SEE REQUIREMENT 6

USER ACTIVITIES
ARMY-MI, ME
NAVY-MC, OS

REVIEWER ACTIVITIES
ARMY-EA
AIR FORCE-82
DLA-IS
NSA-NS

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REVISED

APPROVED 1 AUG 84

P A AR Other Cust SH 99	INTERNATIONAL INTEREST	TITLE SCREW, CAP, HEXAGON HEAD (FINISHED HEXAGON BOLT), STEEL, GRADE 5 UNC-2A	**MILITARY STANDARD** **MS51491**	
PROCUREMENT SPECIFICATION FF-S-85	SUPERSEDES		PAGE 1 OF 2	

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USER ACTIVITIES
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REVIEWER ACTIVITIES
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REQUIREMENTS·

1. MATERIAL MEDIUM CARBON STEEL, GRADE 5, IN ACCORDANCE WITH PROCUREMENT SPECIFICATION

2. FINISH. ZINC COATING, IN ACCORDANCE WITH ASTM B633-78, Fe/Z_n5 (5 µm), TYPE II

3. THREADS. THE THREADS SHALL BE IN ACCORDANCE WITH FED-STD-H28/2. ACCEPTABILITY OF SCREW THREADS SHALL BE IN ACCORDANCE WITH FED-STD-H28/20, SYSTEM 21.

4. THREAD LENGTH. MINIMUM THREAD LENGTH SHALL BE TWICE THE DIAMETER PLUS .250 INCH THE TOLERANCE SHALL BE PLUS .188 INCH OR 2-1/2 THREADS, WHICHEVER IS GREATER. ON SCREWS THAT ARE TOO SHORT FOR MINIMUM THREAD LENGTHS, THE DISTANCE FROM THE BEARING SURFACE OF THE HEAD TO THE FIRST COMPLETE THREAD SHALL NOT EXCEED THE LENGTH OF 2-1/2 THREADS FOR SIZES UP TO AND INCLUDING 1 000 INCH AND 3-1/2 THREADS FOR SIZES LARGER THAN 1.000 INCH.

5. HEAD MARKING. HEADS SHALL BE MARKED IN ACCORDANCE WITH MIL-HDBK-131. RAISED MARKINGS ARE PREFERRED, STAMPINGS ARE ACCEPTABLE.

6. BASED ON MINIMUM ULTIMATE TENSILE STRENGTH OF 120,000 PSI FOR DIAMETERS THRU 1.000 INCH, 105,000 PSI FOR 1 125 INCH THRU 1 500 INCH DIAMETERS TENSILE LOADS ARE CALCULATED BY THE STRESS AREAS INDICATED IN FED-STD-H28/2

7. PART NUMBERS. THE BASIC MS PART NUMBER IS FOLLOWED BY A DASH NUMBER FROM TABLE I.

EXAMPLE: MS51491-001 - SCREW, CAP, HEXAGON HEAD, STEEL, GRADE 5, .250-20UNC-2A,
LENGTH 875, ZINC COATED

NOTES·

1 ALL DIMENSIONS ARE IN INCHES.

2. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BIDS OR REQUEST FOR PROPOSAL, EXCEPT THAT REFERENCED ADOPTED INDUSTRY DOCUMENTS SHALL GIVE THE DATE OF THE ISSUE ADOPTED

3. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN

APPROVED 1 AUG 84 REVISED

P A **AR** Other Cust **SH** **99**	INTERNATIONAL INTEREST	TITLE SCREW,CAP,HEXAGON HEAD (FINISHED HEXAGON BOLT), STEEL,GRADE 5 UNC-2A		**MILITARY STANDARD** **MS51491**
PROCUREMENT SPECIFICATION FF-S-85	SUPERSEDES			PAGE 2 OF 2

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